SMART COMPLAINT MANAGEMENT SYSTEM

COMBINED CODE COVERAGE + BRANCH COVERAGE REPORT

Coverage.py measurement summary

Project detail	Value
Group	8
Course	Software Engineering (UE24CS341A)
Institution	PES University
Version	1.0
Date	11 November 2026

PREPARED BY

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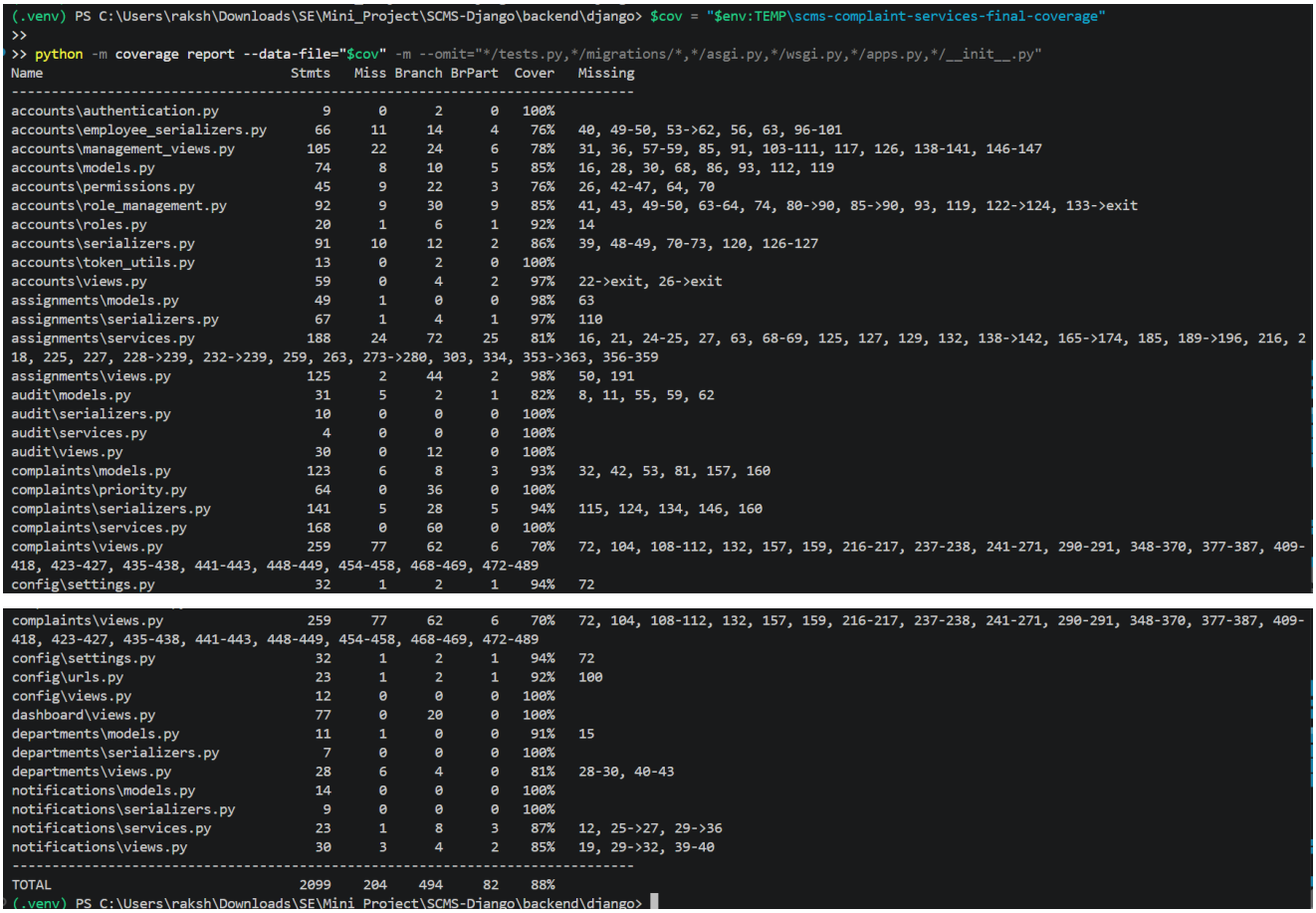
1. Measurement Scope

Coverage was measured with Coverage.py using branch measurement. The source scope includes accounts, assignments, audit, complaints, config, dashboard, departments, and notifications. The filtered baseline report excluded test files, migrations, ASGI/WSGI modules, app configuration files, and package initializers.

2. Latest Overall Result

Metric	Reported result
HTML coverage overview (full report scope)	93% (as shown in the HTML screenshot)
Filtered application report (terminal)	88% (TOTAL row; exclusions applied)
Filtered report statement counts	2,099 statements; 204 missed (90.3% statement coverage)
Filtered report branch counts	494 branch opportunities; 82 partially covered branches (branch measurement enabled)
Coverage.py branch measurement	Enabled (--branch)
Full Django suite in coverage run	103 passed
Failures	0
Configured source scope	accounts, assignments, audit, complaints, config, dashboard, departments, notifications

Interpretation: The HTML overview screenshot reports 93% because it includes the broader Coverage.py report scope, including package/app and migration files. The terminal screenshot reports 88% after excluding tests, migrations, ASGI/WSGI modules, app configuration files, and package initializers. These figures use different report scopes. For the filtered terminal report, 2,099 statements with 204 missed gives 90.3% statement coverage. Branch measurement is enabled; the terminal table reports 494 branch opportunities and 82 partially covered branches, but does not show a separate aggregate branch-only percentage. The combined filtered Coverage.py total is displayed as 88%.

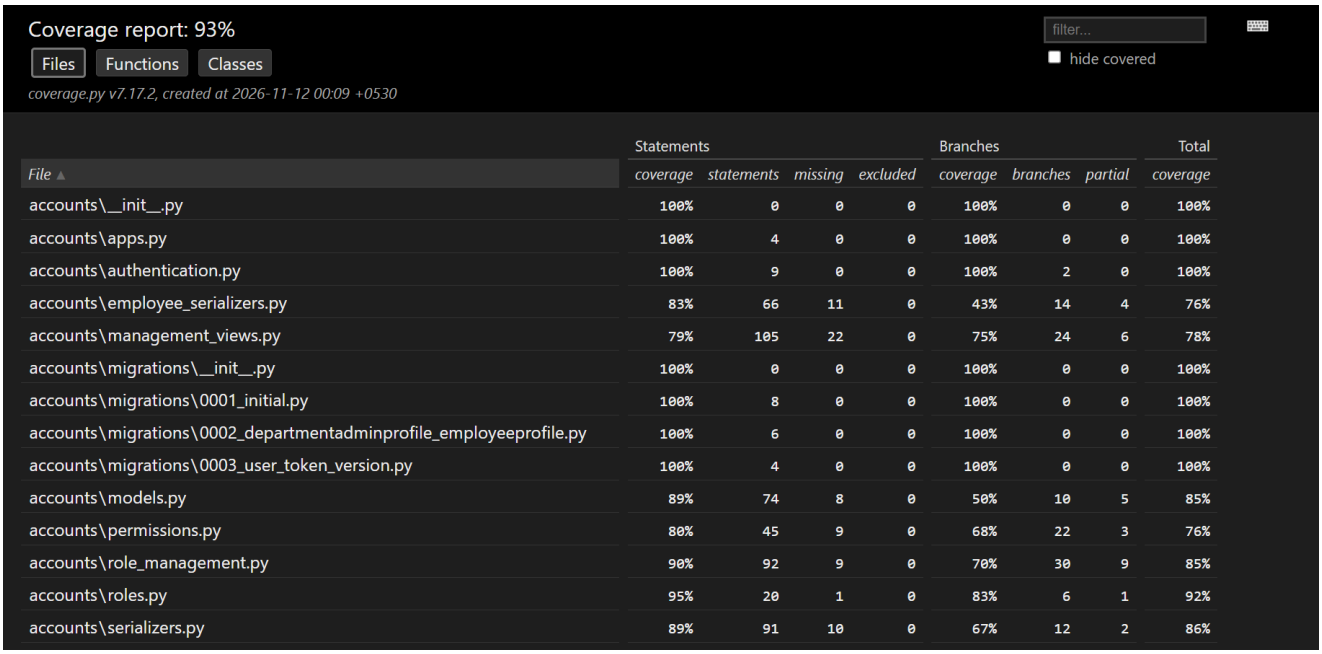


```
(.venv) PS C:\Users\raksh\Downloads\SE\Mini_Project\SCMS-Django\backend\django> $cov = "$env:TEMP\scms-complaint-services-final-coverage"
>>
>> python -m coverage report --data-file="$cov" -m --omit="*/tests.py,*/migrations/*,*/asgi.py,*/wsgi.py,*/apps.py,*/__init__.py"
Name                                  Stmts   Miss Branch BrPart  Cover   Missing
---------------------------------------------------------------------------------
accounts\authentication.py                9      0      2      0   100%
accounts\employee_serializers.py         66     11     14      4    76%   40, 49-50, 53->62, 56, 63, 96-101
accounts\management_views.py            105     22     24      6    78%   31, 36, 57-59, 85, 91, 103-111, 117, 126, 138-141, 146-147
accounts\models.py                       74      8     10      5    85%   16, 28, 30, 68, 86, 93, 112, 119
accounts\permissions.py                  45      9     22      3    76%   26, 42-47, 64, 70
accounts\role_management.py              92      9     30      9    85%   41, 43, 49-50, 63-64, 74, 80->90, 85->90, 93, 119, 122->124, 133->exit
accounts\roles.py                        20      1      6      1    92%   14
accounts\serializers.py                  91     10     12      2    86%   39, 48-49, 70-73, 120, 126-127
accounts\token_utils.py                  13      0      2      0   100%
accounts\views.py                        59      0      4      2    97%   22->exit, 26->exit
assignments\models.py                    49      1      0      0    98%   63
assignments\serializers.py               67      1      4      1    97%   110
assignments\services.py                 188     24     72     25    81%   16, 21, 24-25, 27, 63, 68-69, 125, 127, 129, 132, 138->142, 165->174, 185, 189->196, 216, 2
18, 225, 227, 228->239, 232->239, 259, 263, 273->280, 303, 334, 353->363, 356-359
assignments\views.py                    125      2     44      2    98%   50, 191
audit\models.py                          31      5      2      1    82%   8, 11, 55, 59, 62
audit\serializers.py                     10      0      0      0   100%
audit\services.py                         4      0      0      0   100%
audit\views.py                           30      0     12      0   100%
complaints\models.py                    123      6      8      3    93%   32, 42, 53, 81, 157, 160
complaints\priority.py                   64      0     36      0   100%
complaints\serializers.py               141      5     28      5    94%   115, 124, 134, 146, 160
complaints\services.py                  168      0     60      0   100%
complaints\views.py                     259     77     62      6    70%   72, 104, 108-112, 132, 157, 159, 216-217, 237-238, 241-271, 290-291, 348-370, 377-387, 409-
418, 423-427, 435-438, 441-443, 448-449, 454-458, 468-469, 472-489
config\settings.py                       32      1      2      1    94%   72
```

```
complaints\views.py                     259     77     62      6    70%   72, 104, 108-112, 132, 157, 159, 216-217, 237-238, 241-271, 290-291, 348-370, 377-387, 409-
418, 423-427, 435-438, 441-443, 448-449, 454-458, 468-469, 472-489
config\settings.py                       32      1      2      1    94%   72
config\urls.py                           23      1      2      1    92%   100
config\views.py                          12      0      0      0   100%
dashboard\views.py                       77      0     20      0   100%
departments\models.py                    11      1      0      0    91%   15
departments\serializers.py                7      0      0      0   100%
departments\views.py                     28      6      4      0    81%   28-30, 40-43
notifications\models.py                  14      0      0      0   100%
notifications\serializers.py              9      0      0      0   100%
notifications\services.py                23      1      8      3    87%   12, 25->27, 29->36
notifications\views.py                   30      3      4      2    85%   19, 29->32, 39-40
---------------------------------------------------------------------------------
TOTAL                                  2099    204    494     82    88%
(.venv) PS C:\Users\raksh\Downloads\SE\Mini_Project\SCMS-Django\backend\django>
```

3. Selected Module Coverage

Module	Earlier reported coverage	Latest reported coverage
assignments/views.py	42%	98%
audit/views.py	26%	100%
dashboard/views.py	64%	100%
complaints/priority.py	67.71%	100%
complaints/serializers.py	68.05%	94.08%
complaints/services.py	73.25%	100%



Coverage report: 93%

Files | Functions | Classes

filter... | hide covered

coverage.py v7.17.2, created at 2026-11-12 00:09 +0530

	Statements				Branches			Total
File ▲	coverage	statements	missing	excluded	coverage	branches	partial	coverage
accounts__init__.py	100%	0	0	0	100%	0	0	100%
accounts\apps.py	100%	4	0	0	100%	0	0	100%
accounts\authentication.py	100%	9	0	0	100%	2	0	100%
accounts\employee_serializers.py	83%	66	11	0	43%	14	4	76%
accounts\management_views.py	79%	105	22	0	75%	24	6	78%
accounts\migrations__init__.py	100%	0	0	0	100%	0	0	100%
accounts\migrations\0001_initial.py	100%	8	0	0	100%	0	0	100%
accounts\migrations\0002_departmentadminprofile_employeeprofile.py	100%	6	0	0	100%	0	0	100%
accounts\migrations\0003_user_token_version.py	100%	4	0	0	100%	0	0	100%
accounts\models.py	89%	74	8	0	50%	10	5	85%
accounts\permissions.py	80%	45	9	0	68%	22	3	76%
accounts\role_management.py	90%	92	9	0	70%	30	9	85%
accounts\roles.py	95%	20	1	0	83%	6	1	92%
accounts\serializers.py	89%	91	10	0	67%	12	2	86%

4. Coverage Improvements

Assignment tests cover ownership, department scope, employee workflow actions, reassignment, workload access, and evidence validation.

Audit tests cover role-based access, department filters, actor visibility, ownership, invalid roles, and profileless-admin behavior.

Dashboard tests cover authentication failures, role-specific responses, ownership, department scoping, empty data, and missing employee records.

Priority tests cover thresholds, explicit zero and positive affected counts, text fallbacks, combined factors, and safety triggers.

Serializer tests cover required/optional fields, invalid values, updates, related-object serialization, attachments, and priority-policy updates.

Service tests cover creation, routing, owner restrictions, transitions, resolution, reopening, audit/history, and rollback.

5. Coverage Execution Command

```
python -m coverage run --branch --source
accounts,assignments,audit,complaints,config,dashboard,departments,notifications manage.py test
```

The final supplied run summary records 103 passing Django tests. The same source scope and saved baseline exclusions were used for the reported overall comparison.

6. Interpretation and Limitations

Coverage indicates which instrumented statements and branches were exercised; it does not prove the absence of defects.

Separate aggregate statement-only and branch-only percentages were not present in the final run summary and must be copied from the raw final Coverage.py report if specifically required.

Some serializer inputs are rejected by Django REST Framework field validation before custom validators execute; invalid-input behavior is covered even where those custom lines remain uncovered.

assignments/views.py was reported at 98%; selected high-value workflow modules reached 100%.

7. Conclusion

The supplied branch-enabled Coverage.py evidence shows 93% in the full HTML overview and 88% in the filtered terminal report. Within the filtered report, statement coverage is 90.3% (2,099 statements, 204 missed). Branch measurement is enabled, with 494 branch opportunities and 82 partially covered branches; a separate aggregate branch-only percentage is not available in the supplied terminal output. The full Django suite completed with 103 passing tests. These figures are reported with their respective scopes so the two screenshots can be interpreted correctly.